CS3003: IT Project Management

Surgery 1. Project Triangle/Business Case/Costs

Q1. Project Triangle

Which (one) of the following statements is true?

a. An increase in project scope is likely to increase project cost.
b. A decrease in the project time is likely to increase project quality.
c. An increase in the project quality requirements is likely to decrease project cost.
d. A decrease in the project cost is likely to decrease project time.

Answer

Option a, please consider Project Triangle – if scope is increased, then more resources (i.e. more cost) should be added to deliver the increased scope of the project in the same schedule and with the same, appropriate level of quality.

Q2. Project Triangle

In *Agile* projects, how is the Project Triangle often reinterpreted?

a. Scope is fixed, and time/cost are flexible
b. Time and cost are fixed, scope is flexible
c. All three (scope, cost, time) are fixed in every iteration
d. Triple constraint is not used in agile

Answer

Option b - in Agile projects, iterations (sprints) are time-boxed with fixed cost (Agilel team), so scope is flexible

Q3. Business Case/TCO

Which (one) of the following statements is true?

What is the relationship between TCO and ROI?

a. TCO is used to estimate all costs; ROI compares total benefits to those costs
b. ROI replaces TCO
c. TCO equals ROI × benefits
d. TCO only considers revenue, ROI only costs

Answer

Option a – the rest of the options actually do not make sense

Q4. Business Case/TCO

Your company is about to buy a software package, Package A, that consists of three components, namely A1, A2 and A3. The three components are licensed by the vendor as follows:

- A1 is licensed on a per user basis, costing £50 per User that will have access to the package.
- A2 is licensed on a per user basis, costing £150 per User that will have access to the package.
- A3 is licensed based on the number of CPUs that are installed on the application server that Package A will run, costing £10 000 per installed CPU. It is estimated that to be able to perform adequately in the production environment, Package A requires 2 CPUs for up to 250 Users, 4 CPUs for 251 to 500 Users and 8 CPUs for 501 to 1000 Users.

Moreover, starting from the second year, your company will have to pay the vendor of Package A an annual 20% maintenance fee over the license fee. Finally, each CPU of the production environment costs £5000.

If variable N denotes the number of Users and variable M the number of CPUs, then, based on the previous facts:

a) devise the formulas to calculate the 5-year Total Cost of Ownership (TCO) of the investment that your company would have to make for Package A. Based on the previous formulas, calculate the 5-year TCO of Product A for 300 Users.

b) If the company estimates that as a result of the deployment of Product A, it will achieve an annual profit of £60 000, what is the Payback Period of the investment on Product A?

Notes: There will be <u>no</u> disaster site for Product A. There is <u>no</u> maintenance fee for Hardware (CPUs). The time-value of money should <u>not</u> be taken into account for the TCO calculation. <u>No other cost factors</u> should be taken into account in calculating the Payback Period.

Sample Answer

If N denotes the number of users and M the number of CPUs then the 5-year TCO is given by the following formulae:

TCO = Acquisition Cost (or CAPEX) + Maintenance Cost (or OPEX) =($Cost_{licensing\ SW}$ + $Cost_{CPUs}$) + $Cost_{SW\ Maintenance\ (5\text{-}years)}$ + $Cost_{HW\ Maintenance\ (5\text{-}years)}$

- $Cost_{licensing\ SW}$ = 50*N + 150*N + 10000*M with M=2 (0<N<=250), M=4 (250<N<=500), M=8 (500<N<=1000).
- $Cost_{CPUs}$ = 5000*M with M=2 (0<N<=250), M=4 (250<N<=500), M=8 (500<N<=1000).
- $Cost_{SW\ Maintenance\ (5\text{-}years)}$ = 4*[20% ($Cost_{licensing}$)]
- $Cost_{HW\ Maintenance\ (5\text{-}years)}$ = 0;

If N = 300 then M = 4 and

- $Cost_{licensing}$ = 50*300 + 150*300 + 10000*4 = 15000+45000+40 000 = 100 000
- $Cost_{CPUs}$ = 5000*4 = 20 000

- Cost $_{\text{Maintenance (5-years)}}$ = 4*[20% (100 000)] = 4*20 000 = 80 000

Hence, 5-year TCO for 300 Users = 100 000 + 20 000 + 80 000 = 200 000.

In what concerns the PP, the CAPEX cost of the investment is Cost $_{\text{licensing}}$ + Cost $_{\text{CPUs}}$ = 100 000+20 000 = 120 000. Hence, PP = 120 000 / 60 000 = 2 years. Note. The calculation above ignores OPEX costs in the PP calculation (i.e. takes into account the initial investment only).

Q5. Business Case/NPV

Complete the following table by calculating the NPV valuation of two projects, Project 1 and Project 2. The discount rate is 10% and discounting starts in Year 1. What conclusions can be reached by comparing the two projects?

Answer

Project 1						
	YEAR 1	YEAR 2	YEAR 3	YEAR 4	YEAR 5	TOTALS
Benefits	0	2000	3000	4000	5000	14000
Costs	5000	1000	1000	1000	1000	9000
Net Cash Flow (NCF)	(5000)	1000	2000	3000	4000	5000
Discount Factor	0.909	0.826	0.751	0.683	0.621	
NCF x DF	-4545	826	1502	2049	2484	
NPV	**2316**					
Project 2						
	YEAR 1	YEAR 2	YEAR 3	YEAR 4	YEAR 5	TOTALS
Benefits	1000	2000	4000	4000	4000	15000
Costs	2000	2000	2000	2000	2000	10000
Net Cash Flow	(1000)	0	2000	2000	2000	5000
Discount Factor	0.909	0.826	0.751	0.683	0.621	
NCF x DF	-909	0	1502	1366	1242	
NPV	**3201**					

Main conclusion: despite the fact that the two projects have an equal Total Net Cash Flow of 5K, Project No 2 has a better NPV, hence financially it makes more sense than Project 1.

Q6. Project Cost Estimation

Which (one) of the following statements is true?

The accuracy of an estimate should:

a. decrease as a project progresses through its life cycle.

b. increase as a project progresses through its life cycle.

c. stay constant throughout the project life cycle.

d. vary independently of where the project is in its life cycle.

Answer

Option b – as a project progresses, more facts and details are available (e.g. requirements), hence estimates should become more accurate.

Q7. Project Cost Estimation

A project manager is estimating the effort of a design activity. The team provides the following estimates:

Optimistic (O): 5 person days, **Most Probable (MP):** 8 person days, **Pessimistic (P):** 17 person days

Using the **PERT three-point estimate formula**, what is the expected total effort of the design activity?

a. 9 person days
b. 10 person days
c. 12 person days
d. 17 person days

Answer

Option a, 9 person days – using the PERT formula, **E = (4*MP + O + P) / 6** => E = (4*8 + 5 + 17)/6 => E=54/6 => E = 9

Q8. CAPEX/OPEX

Which (one) of the following is an example of **CAPEX**?

a. Annual server hosting fees
b. Buying a new piece of hardware
c. Monthly software subscription
d. Employee training travel costs

Answer

Option b - buying a new piece of hardware

Q9. CAPEX/OPEX

A project manager is evaluating two options:

- **Option A:** Buy servers for £500,000 + £50,000 annual maintenance
- **Option B:** Use cloud services at £120,000 per year

If the company wants to avoid large CAPEX investments, which option is better?

a. Option A
b. Option B
c. Both equal
d. Cannot be determined

Answer

Option b – option a includes £500,000 of CAPEX cost, option b is OPEX only, thus avoiding upfront CAPEX

Q10. CAPEX/OPEX

During a project selection meeting, one project involves high CAPEX, while another primarily incurs OPEX. Which (one) of the following statements is most accurate?

a. OPEX-heavy projects have no long-term financial impact
b. CAPEX-heavy projects tie up cash upfront but may reduce OPEX later
c. OPEX-heavy projects always have higher ROI
d. CAPEX and OPEX classifications do not matter in project selection

Answer

Option b – please reflect on the reasons the rest of options are incorrect.